
MA 578: BAYESIAN STATISTICS

Chapter 8: Hierarchical Models and Empirical Bayes

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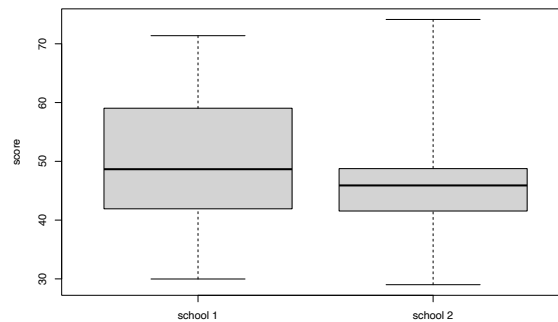
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1 Hierarchical Models

- Suppose we have the following data for math scores for 10th grade students from two schools
 School 1 sample size: 31, school 1 total number of 10th graders: 600
 School 2 sample size: 28, school 2 total number of 10th graders: 600



θ_1 : Average score of all 600 10th graders in school 1.

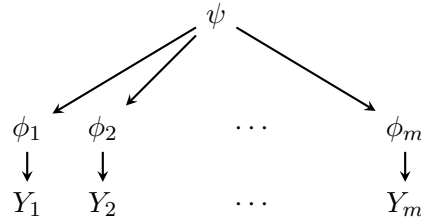
θ_2 : Average score of all 600 10th graders in school 2.

- We are interested in selecting a model for the data
- Two extreme models:
 - Two groups have completely distinct distributions
 $\Rightarrow$ We do analysis for the two groups separately
 - Two groups have exactly identical distributions
 $\Rightarrow$ We pool the data and assume $\theta_1 = \theta_2$
- The first option ignores shared information and the second option ignores differences between the two groups
- Since the two samples are measuring the same thing, it makes sense to use some of the information in one group to help estimate the mean in the other group
- Bayesian analysis based on a hierarchical modeling can be used to allow for such information sharing across the groups

Hierarchical Model

- The data in the example above includes
 - population of schools
 - population of students within each school

- Examples:
 - Students within several schools
 - Patients within several hospitals
 - People within regions within countries
- Datasets with such a hierarchy of nested populations are called *hierarchical*
- A two level hierarchical data contains
 - Level one: Groups (j)
 - Level two: Units within groups $y_{i,j}$
- Let $\{\phi_j\}$ denote group-specific parameters



- We assume the following on our data, which results from two levels of exchangeability assumption, followed by double application of de Finetti's theorem

$$\begin{array}{ll}
 y_{1,j}, \dots, y_{n_j,j} \mid \phi_j \stackrel{iid}{\sim} p(y \mid \phi_j) & \text{within-group sampling model} \\
 \phi_1, \dots, \phi_m \mid \psi \stackrel{iid}{\sim} p(\phi \mid \psi) & \text{between-group sampling model} \\
 \psi \sim p(\psi) & \text{prior distribution}
 \end{array}$$

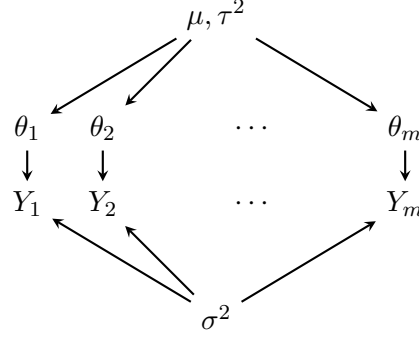
- $p(y \mid \phi)$ and $p(\phi \mid \psi)$ represent sampling variability
- $p(\psi)$ represents information (belief) about an unknown quantity

1.1 Hierarchical Normal Model

- The within-group and between-group sampling models are both normal

$$\begin{aligned}
 \phi_j &= \{\theta_j, \sigma^2\}, & p(y \mid \phi_j) &= \mathcal{N}(\theta_j, \sigma^2) \\
 \psi &= \{\mu, \tau^2\}, & p(\theta_j \mid \psi) &= \mathcal{N}(\mu, \tau^2)
 \end{aligned}$$

- For now, we assume $p(\phi \mid \psi)$ only describes the heterogeneity across group means, not any heterogeneity in group-specific variances



- Unknown parameters in this model: μ, τ^2, σ^2
- We use the following priors on these parameters

$$\sigma^2 \sim \text{Inv-Gamma}\left(\frac{\nu_0}{2}, \frac{\nu_0 \sigma_0^2}{2}\right)$$

$$\tau^2 \sim \text{Inv-Gamma}\left(\frac{\eta_0}{2}, \frac{\eta_0 \tau_0^2}{2}\right)$$

$$\mu \sim \mathcal{N}(\mu_0, \gamma_0^2)$$

1.2 Posterior Inference

- Unknown quantities:
 - Group-specific mean: $\{\theta_1, \dots, \theta_m\}$
 - Within-group sampling variability: σ^2
 - Mean and variance of the distribution of group-specific means: (μ, τ^2)
- Our goal is to do joint posterior inference on the parameters
- We use a Gibbs sampler to approximate the joint posterior distribution

$$p(\theta_1, \dots, \theta_m, \mu, \tau^2, \sigma^2 \mid y_1, \dots, y_m)$$

- We use the following factorization:

$$\begin{aligned}
 & p(\theta_1, \dots, \theta_m, \mu, \tau^2, \sigma^2 \mid y_1, \dots, y_m) \\
 & \propto p(\mu, \tau^2, \sigma^2) p(\theta_1, \dots, \theta_m \mid \mu, \tau^2, \sigma^2) p(y_1, \dots, y_m \mid \theta_1, \dots, \theta_m, \mu, \tau^2, \sigma^2) \\
 & \propto p(\mu) p(\tau^2) p(\sigma^2) \left\{ \prod_{j=1}^m p(\theta_j \mid \mu, \tau^2) \right\} \left\{ \prod_{j=1}^m \prod_{i=1}^{n_j} p(y_{i,j} \mid \theta_j, \sigma^2) \right\}
 \end{aligned}$$

- Note that given (θ_j, σ^2) , Y_j is independent of (μ, τ^2) .

Full conditional distributions:

- Regarding (μ, τ^2) , collecting the terms in the factorization relevant to (μ, τ^2) ,

$$p(\mu \mid \theta_1, \dots, \theta_m, \tau^2, \sigma^2, y_1, \dots, y_m) \propto p(\mu) \prod p(\theta_j \mid \mu, \tau^2)$$

$$p(\tau^2 \mid \theta_1, \dots, \theta_m, \mu, \sigma^2, y_1, \dots, y_m) \propto p(\tau^2) \prod p(\theta_j \mid \mu, \tau^2)$$

- Comparing the setting here with what we had in one-sample normal model, $\{\theta_1, \dots, \theta_m\}$ is playing the role of $\{y_1, \dots, y_m\}$ in that setting, and (μ, τ^2) is playing the role of (θ, σ^2) in that setting.
- Therefore, applying the results from before,

$$p(\mu \mid \theta_1, \dots, \theta_m, \tau^2, \sigma^2) = \mathcal{N}\left(\frac{m\bar{\theta}/\tau^2 + \mu_0/\gamma_0^2}{m/\tau^2 + 1/\gamma_0^2}, \frac{1}{m/\tau^2 + 1/\gamma_0^2}\right)$$

$$p(\tau^2 \mid \theta_1, \dots, \theta_m, \mu, \sigma^2) = \text{Inv-Gamma}\left(\frac{\eta_0 + m}{2}, \frac{\eta_0\tau_0^2 + \sum(\theta_j - \mu)^2}{2}\right)$$

- Regarding θ_j , collecting the terms in the factorization relevant to θ_j ,

$$p(\theta_j \mid \mu, \tau^2, \sigma^2, y_1, \dots, y_m) \propto p(\theta_j \mid \mu, \tau^2) \prod_{i=1}^{n_j} p(y_{i,j} \mid \theta_j, \sigma^2)$$

- This is exactly the same setup as the one-sample normal model, in which $p(\theta_j \mid \mu, \tau^2)$ is the prior distribution instead of the sampling model for the θ 's. Therefore,

$$p(\theta_j \mid y_{1,j}, \dots, y_{n_j,j}, \sigma^2) = \mathcal{N}\left(\frac{n_j\bar{y}_j/\sigma^2 + \mu/\tau^2}{n_j/\sigma^2 + 1/\tau^2}, \frac{1}{n_j/\sigma^2 + 1/\tau^2}\right)$$

- Regarding σ^2 , collecting the terms in the factorization relevant to σ^2 ,

$$p(\sigma^2 \mid \theta_1, \dots, \theta_m, y_1, \dots, y_m) \propto p(\sigma^2) \prod_{j=1}^m \prod_{i=1}^{n_j} p(y_{i,j} \mid \theta_j, \sigma^2)$$

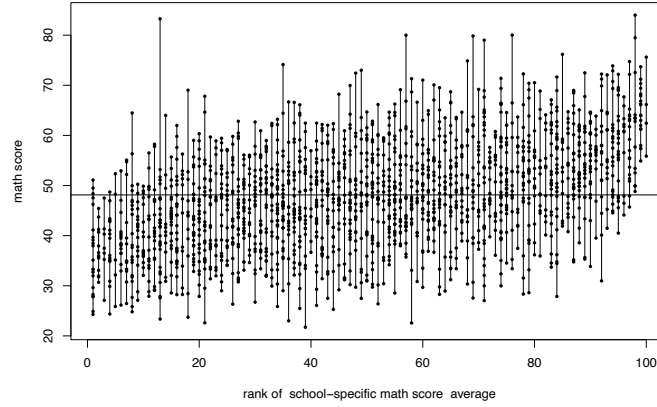
$$\propto (\sigma^2)^{-\nu_0/2+1} e^{-\frac{\nu_0\sigma_0^2}{2\sigma^2}} (\sigma^2)^{-\sum n_j/2} e^{-\frac{\sum \sum (y_{i,j} - \theta_j)^2}{2\sigma^2}}$$

- This is similar to that in the one-sample normal model, except now we have information about σ^2 from m separate groups. We conclude

$$p(\sigma^2 \mid \theta, y_1, \dots, y_m) = \text{Inv-Gamma}\left(\frac{1}{2}[\nu_0 + \sum_{j=1}^m n_j], \frac{1}{2}[\nu_0\sigma_0^2 + \sum_{j=1}^m \sum_{i=1}^{n_j} (y_{i,j} - \theta_j)^2]\right)$$

- Note that $\sum_{j=1}^m \sum_{i=1}^{n_j} (y_{i,j} - \theta_j)^2$ is the sum of squared residuals across all groups, conditioned on the within-group means $\Rightarrow$ the distribution is concentrating around a pooled-sample estimate of the variance.

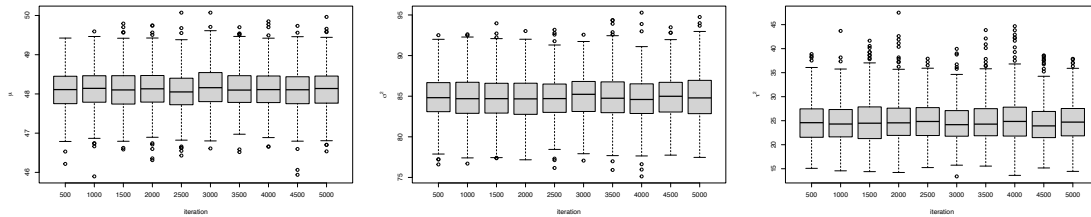
Example. Going back to the math scores example, the figure below shows the scores for 100 schools:



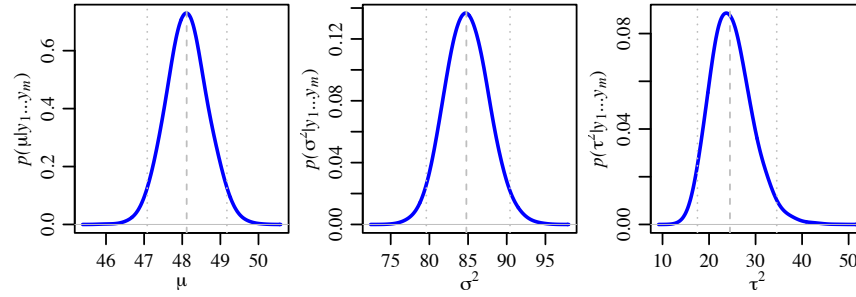
- Prior parameters we need to specify are
 - (ν_0, σ_0^2) for $p(\sigma^2)$
 - (η_0, τ_0^2) for $p(\tau^2)$
 - (μ_0, γ_0^2) for $p(\mu)$
- We choose $\sigma_0^2 = 100, \nu_0 = 1, \tau_0^2 = 100, \eta_0 = 1, \mu_0 = 50, \gamma_0^2 = 25$
- Posterior approximation proceeds by iterative sampling:
 - sample $\mu^{(s+1)} \sim p(\mu \mid \theta_1^{(s)}, \dots, \theta_m^{(s)}, \tau^{2(s)})$
 - sample $\tau^{2(s+1)} \sim p(\tau^2 \mid \theta_1^{(s)}, \dots, \theta_m^{(s)}, \mu^{(s+1)})$
 - sample $\sigma^{2(s+1)} \sim p(\sigma^2 \mid \theta_1^{(s)}, \dots, \theta_m^{(s)}, y_1, \dots, y_m)$
 - for each $j \in \{1, \dots, m\}$, sample $\theta_j^{(s+1)} \sim p(\theta_j \mid \mu^{(s+1)}, \tau^{2(s+1)}, \sigma^{2(s+1)}, y_j)$

MCMC diagnostics:

First, we look at the trace plots, or in this case, since we are using a large number for iterations and it is hard to read the trace, we look at the box plot of sequential groups of samples, seen in the figure below.



Now that we are rather confident about the convergence of the chains, we look at the posterior densities of μ, σ^2, τ^2 :

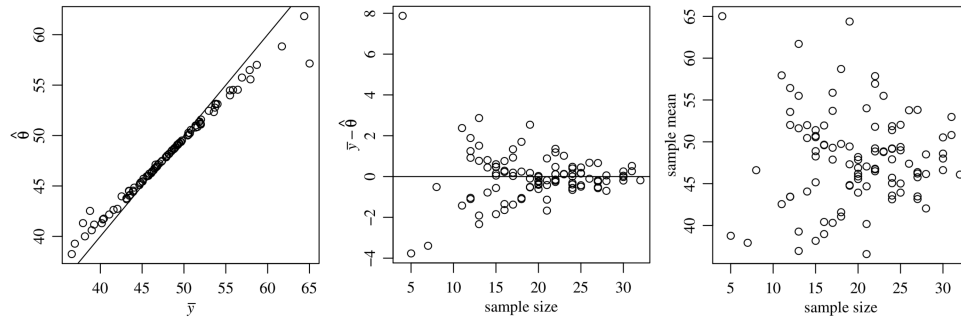


- One of the motivations behind hierarchical modeling is that information can be shared across groups.
- Recall that

$$\mathbb{E}[\theta_j \mid y_j, \mu, \tau, \sigma] = \frac{n_j \bar{y}_j / \sigma^2 + \mu / \tau^2}{n_j / \sigma^2 + 1 / \tau^2}$$

- That is, the expected value of θ_j is pulled a bit from $\bar{y}_j$ towards μ by an amount depending on n_j .
- This effect is called *shrinkage*.
- Figure below

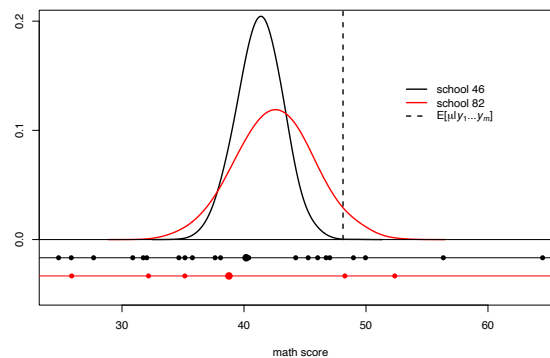
- Plots $\bar{y}_j$ versus $\hat{\theta}_j = \mathbb{E}[\theta_j \mid y_1, \dots, y_m]$ for each group to see the shrinkage.
- Shows the amount of shrinkage as a function of the group-specific sample size, indicating that groups with low sample sizes get shrunk the most (why?)
- Shows that extreme sample averages tend to be associated with schools with small sample sizes (why?)



- Suppose we would like to rank the schools. We consider two options: Rank the schools according to

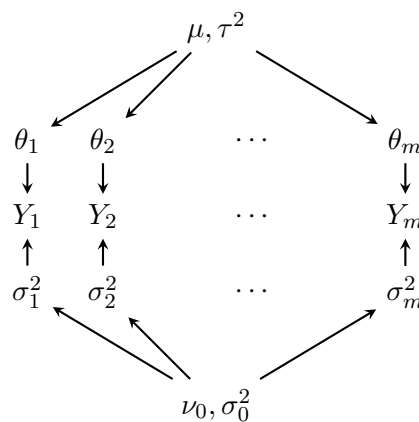
- school-specific posterior expectations $\hat{\theta}_j = \mathbb{E}[\theta_j \mid y_1, \dots, y_m]$

- school-specific sample averages $\bar{y}_j$
- Consider the posterior distributions of θ_{46} and θ_{82} , in the figure below.
Dotplots below the posterior densities are raw data. Note that school 82 has only 5 data points, hence the larger variance
Large dots representing the sample means $\bar{y}_{46}$ and $\bar{y}_{82}$
- Note that the order for posterior expectations and sample averages is reverse
- **Question.** Can you explain why it makes sense to shrink the expectation of school 82 towards the population expectation $\mathbb{E}[\mu \mid y_1, \dots, y_m] = 48.11$ by a greater amount than for the expectation of school 46?
- Hierarchical model reflects the objective fact that there is more evidence that θ_{46} is exceptionally low than there is evidence that θ_{82} is exceptionally low.



1.3 Hierarchical Modeling of Means and Variances

- The population variances can also vary across groups



- $\{(\theta_1, \sigma_1^2), \dots, (\theta_m, \sigma_m^2)\}$: represent within-group sampling distribution
- $\{\mu, \tau^2\}$: represent across-group heterogeneity in means
- $\{\nu_0, \sigma_0^2\}$: represent across-group heterogeneity in variances
- Sampling model:

$$Y_{1,j}, \dots, Y_{n_j,j} \stackrel{iid}{\sim} \mathcal{N}(\theta_j, \sigma_j^2)$$

- Full conditional distribution for each θ_j

$$p(\theta_j | y_j, \mu, \tau, \sigma_j^2) = \mathcal{N}\left(\frac{n_j \bar{y}_j / \sigma_j^2 + \mu / \tau^2}{n_j / \sigma_j^2 + 1 / \tau^2}, \frac{1}{n_j / \sigma_j^2 + 1 / \tau^2}\right)$$

- Sampling model for σ^2 , i.e., sampling model for across-group heterogeneity in population variances:

$$\sigma_1^2, \dots, \sigma_m^2 \stackrel{iid}{\sim} \text{Inv-Gamma}(\nu_0/2, \nu_0 \sigma_0^2/2)$$

- Therefore, as seen before, full conditional distribution for each σ_j^2

$$p(\sigma_j^2 | y_j, \theta_j) = \text{Inv-Gamma}([\nu_0 + n_j]/2, [\nu_0 \sigma_0^2 + \sum (y_{i,j} - \theta_j)^2]/2)$$

- Hence, we can again proceed by using a Gibbs sampler.
- We need to
 - specify the prior distributions for ν_0 and σ_0^2
 - obtain their full conditional distributions
- For σ_0^2 , if $p(\sigma_0^2) \sim \text{Gamma}(a, b)$

$$p(\sigma_0^2 | \sigma_1^2, \dots, \sigma_m^2, \nu_0) \sim \text{Gamma}(a + \frac{1}{2}m\nu_0, b + \frac{1}{2}\nu_0 \sum_{j=1}^m (1/\sigma_j^2))$$

- For ν_0 , a simple conjugate prior does not exist. However, if we restrict ν_0 to be a whole number, then we can obtain the full conditional distribution:
- For example, if we choose the prior to be geometric so that $p(\nu_0) \propto e^{-\alpha\nu_0}$, then for the full conditional distribution we have

$$\begin{aligned} p(\nu_0 | \sigma_1^2, \dots, \sigma_m^2, \sigma_0^2) &\propto p(\nu_0) p(\sigma_1^2, \dots, \sigma_m^2 | \nu_0, \sigma_0^2) \\ &\propto \left(\frac{(\nu_0 \sigma_0^2/2)^{\nu_0/2}}{\Gamma(\nu_0/2)} \right)^m \left(\prod_{j=1}^m \frac{1}{\sigma_j^2} \right)^{\nu_0/2-1} \exp\{-\nu_0(\alpha + \frac{1}{2}\sigma_0^2 \sum (1/\sigma_j^2))\} \end{aligned}$$

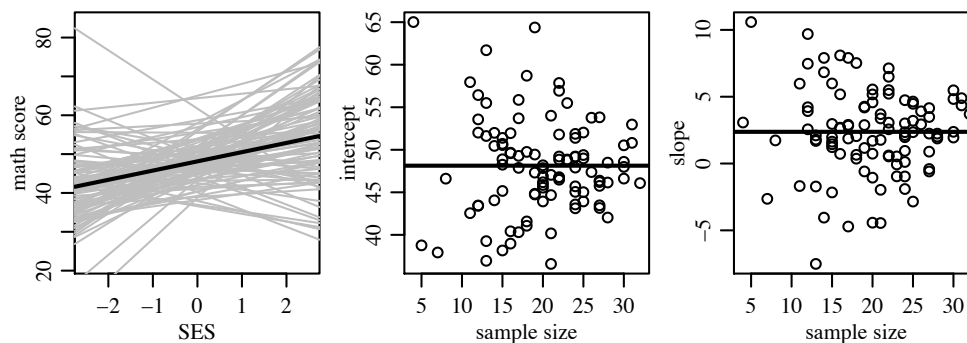
2 Hierarchical Linear Regression Model

- One-sample regression model: Association of the outcome to the covariate
- What if we have more than one group and the said association may differ across groups?

→ Hierarchical regression model

In this model, we have

- Within-group sampling model → Capturing within-group variation: We use regression model for this
- Between-group sampling model → Capturing heterogeneity among regression coefficients across groups
- Recall the math score data.
- Previously, we only looked at expected math scores
- Here, we consider the association between math score and socioeconomic status (SES) (calculated from parental income and education levels)
- We don't want to assume that the association between math scores and SES is the same in all schools and hence, we don't want to just pool the data
- However, we also want to be able to share information across schools to prevent results due to small and non-representative sample
- In figure below, we fit an OLS for each school
Notice that
 - Schools with higher sample size have regression coefficients that are generally close to average
 - Extreme coefficients generally belong to schools with low sample size



- **Side note:** Center the covariates to make the parameters more interpretable
- We **stabilize** the estimates for small sample size groups by **sharing information** across groups using a **hierarchical model**

Normal Hierarchical Linear Regression Model

- Within-group sampling model:

$$Y_{i,j} = \beta_j^\top \mathbf{x}_{i,j} + \epsilon_{i,j}, \quad \{\epsilon_{i,j}\} \stackrel{iid}{\sim} \mathcal{N}(0, \sigma^2)$$

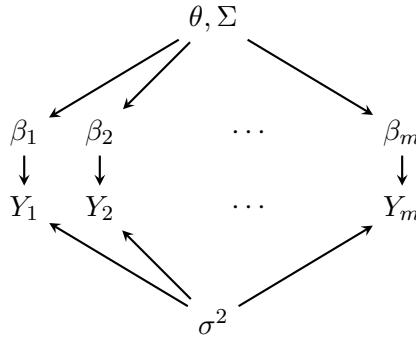
Matrix representation:

$$Y_j \sim \mathcal{N}(X_j \beta, \sigma^2 I)$$

- Group specific data vectors $Y_1, \dots, Y_m$ are conditionally independent given $\beta_1, \dots, \beta_m, \sigma^2$
- Between-group sampling model:

$$\beta_1, \dots, \beta_m \stackrel{iid}{\sim} \mathcal{N}(\theta, \Sigma)$$

- Graphical representation of the model:



- The multivariate normal distribution for $\beta_1, \dots, \beta_m$
 - is not a prior distribution representing uncertainty about one unknown quantity
 - Rather, it is a sampling model representing heterogeneity among a collection of objects
- θ, Σ are unknown parameters to be estimated

Remark. Hierarchical linear regression model is also called *linear mixed effects model*, due to the following alternative parametrization:

Between-group sampling model:

$$\beta_j = \theta + \gamma_j, \quad \gamma_1, \dots, \gamma_m \stackrel{iid}{\sim} \mathcal{N}(0, \Sigma)$$

Therefore,

$$\begin{aligned} Y_{i,j} &= \beta_j^\top \mathbf{x}_{i,j} + \epsilon_{i,j} \\ &= \theta^\top \mathbf{x}_{i,j} + \gamma_j^\top \mathbf{x}_{i,j} + \epsilon_{i,j} \end{aligned}$$

Matrix form:

$$Y_j = \theta^\top X_j + \gamma_j^\top X_j + \epsilon_j$$

θ : fixed effect $\rightarrow$ constant across groups

$\gamma_1, \dots, \gamma_m$: random effects $\rightarrow$ vary across groups

We have both fixed effect and random effect

$\Rightarrow$ mixed effect model

The more general version of mixed effect linear model:

$$Y_{i,j} = \theta^\top \mathbf{x}_{i,j} + \gamma_j^\top \mathbf{z}_{i,j} + \epsilon_{i,j}$$

That is, the regressors corresponding to the fixed and random effects need not be the same. $\mathbf{x}_{i,j}$ and $\mathbf{z}_{i,j}$ may or may not contain overlapping variables.

2.1 Posterior Inference

- We use semi-conjugate prior distributions for θ, Σ, σ^2 :

Prior distributions:

$$\theta \sim \mathcal{N}(\mu_0, \Lambda_0)$$

$$\Sigma \sim \text{Inv-Wishart}(\eta_0, S_0)$$

$$\sigma^2 \sim \text{Inv-Gamma}(\nu_0/2, \nu_0\sigma_0^2/2)$$

- This choice will facilitate obtaining the full conditional distributions for the parameters.
- Therefore, we can use Gibbs sampling to approximate the posterior distributions.

Full conditional distributions

Regarding $\beta_1, \dots, \beta_m$:

- Conditioned on σ^2, θ, Σ , coefficients $\beta_1, \dots, \beta_m$ are independent
- Therefore, for any β_j , the model is the same as the one-sample regression model in which $p(\beta_j \mid \theta, \Sigma)$ is the prior distribution instead of the sampling model for β 's
- Therefore, using the results from Chapter 6, $\beta_j \mid y_j, X_j, \sigma^2, \theta, \Sigma$ has multivariate normal distribution with the following mean and covariance matrix:

$$\begin{aligned} \text{var}(\beta_j \mid y_j, X_j, \sigma^2, \theta, \Sigma) &= (\Sigma^{-1} + X_j^\top X_j / \sigma^2)^{-1} \\ \mathbb{E}[\beta_j \mid y_j, X_j, \sigma^2, \theta, \Sigma] &= (\Sigma^{-1} + X_j^\top X_j / \sigma^2)^{-1} (\Sigma^{-1} \theta + X_j^\top y_j / \sigma^2) \end{aligned}$$

Regarding θ, Σ :

- We need to do inference about the mean and variance of a multivariate normal population
- From Chapter 5,
 - Full conditional distribution for population mean

$$\theta \mid \beta_1, \dots, \beta_m, \Sigma \sim \mathcal{N}(\mu_m, \Lambda_m)$$

$$\text{where } \Lambda_m = (\Lambda_0^{-1} + m\Sigma^{-1})^{-1}$$

$$\mu_m = \Lambda_m(\Lambda_0^{-1}\mu_0 + m\Sigma^{-1}\bar{\beta})$$

$$\text{where } \bar{\beta} = \frac{1}{m} \sum \beta_j$$

- Full conditional distribution for covariance matrix

$$\Sigma \mid \beta_1, \dots, \beta_m, \theta \sim \text{Inv-Wishart}(\eta_0 + m, [S_0 + S_\theta])$$

$$\text{where } S_\theta = \sum_{j=1}^m (\beta_j - \theta)(\beta_j - \theta)^\top$$

Regarding σ^2 :

- σ^2 is common across groups. Therefore, the analysis is similar to the one-sample model, except that data is pooled from all groups:

$$\sigma^2 \mid y, X, \beta \sim \text{Inv-Gamma}([\nu_0 + \sum n_j]/2, [\nu_0\sigma_0^2 + SSR]/2)$$

$$\text{where } SSR = \sum_{j=1}^m \sum_{i=1}^{n_j} (y_{i,j} - \beta_j^\top \mathbf{x}_{i,j})^2$$

Back to the math score data

We take

μ_0 = sample mean of OLS estimator

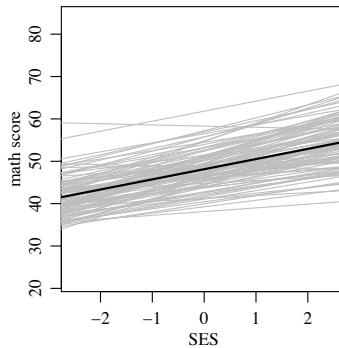
$\Lambda_0 = S_0$ = sample covariance matrix of OLS estimator

$\eta_0 = p + 2 = 4$

σ_0^2 = average of within-group sample variances

$\nu_0 = 1$

Posterior expectations of the 100 school-specific regression lines, and posterior mean of θ :



Notice how the hierarchical model shares information across groups and shrinks extreme regression lines towards the across-group average.

Remark 1. Hierarchical models enable the researcher to separate the prior modeling into two parts

- Structural information: For example, assuming that θ_j s are conditionally iid. Therefore,

$$p(\theta_1, \dots, \theta_m \mid \psi) = \prod_{j=1}^m p(\theta_j \mid \psi)$$

- Subjective information: For example, in the form of a prior distribution $p(\psi)$ on ψ

Remark 2. So far, we described hierarchical models merely as a modeling technique suitable for multi-level data. Another use of hierarchical models is regarding justification of prior parameters. We elaborate on this in the following.

Criticisms of the Bayesian Paradigm

1. The subjective aspects of a Bayesian analysis can always be modified to produce conclusions established in advance

Response: The choice of a prior distribution should always be justifiable by the statistician, that is, it must be based on sound (or “repeatable”) arguments

2. The prior information is rarely rich enough to define a prior distribution exactly

Response: We should include this uncertainty in the Bayesian model

- The hierarchical Bayes analysis $\rightarrow$ within the Bayesian paradigm
- The empirical Bayes analysis $\rightarrow$ combines the frequentist and Bayesian paradigms
- Hierarchical model decomposes the prior distribution into several conditional levels of distributions
- Uncertainty at any of these levels is incorporated into additional prior distributions
- The alternative approach is empirical Bayes analysis, discussed in the following section.

3 Empirical Bayes

- Based on the same perception of imprecision over the prior information, but at a more pragmatic level

- This approach considers that it is illusory to try to model this imprecision by several levels of conditional distributions when the first level is already partly or totally unknown!
- Hyperparameters are considered for the prior, which are estimated from the observations and this “estimated prior” is then used as a regular prior in the subsequent inference
- This approach approximates the prior distribution by frequentist methods
- This approach is not a purely Bayesian
- Empirical Bayes analysis occupies an intermediate position between the classical and Bayesian methods
- **Main idea:** Use the marginal distribution $p(y)$ to determine the prior

$$p(y) = \int_{\Theta} p(y | \theta) p(\theta) d\theta$$

- **Example.**

$$Y | \theta \sim \mathcal{N}(\theta, \sigma^2), p(\theta) = \mathcal{N}(\mu_{\pi}, \sigma_{\pi}^2)$$

$$\text{Then, } p(y) = \mathcal{N}(\mu_{\pi}, \sigma^2 + \sigma_{\pi}^2)$$

- We use the data $y_1, \dots, y_n$ to estimate $p(y)$ or some feature thereof

$$y_1, \dots, y_M \rightarrow p(y) \rightarrow p(\theta)$$

- To make use of knowledge about $p(y)$ in selecting a prior, it is usually necessary to restrict consideration to some class of prior distributions Γ .

3.1 Restricted Classes of Priors

Priors of a Given Functional Form

$$\Gamma = \{p : p(\theta) = g(\theta | \xi), \xi \in \Xi\}$$

- Choice of a prior reduces to the choice of $\xi \in \Xi$
- The parameter ξ is usually called a hyperparameter of the prior
- ξ is considered unknown and to be determined from information about the marginal distribution
- **Example:** If we believe normal distributions with positive mean adequately describe the prior, we can choose to have $\Gamma = \{p : p(\theta) = \mathcal{N}(\theta | \mu, \sigma^2), \mu > 0, \sigma^2 > 0\}$.

Priors of a Given Structural Form

- This type of restriction is for when θ is a vector.
- Example:

$$\Gamma = \{p : p(\theta) = \prod_{i=1}^M p_0(\theta_i), p_0 \text{ an arbitrary density}\}$$

- Structural relationships are often combined with the given functional form idea to yield classes of priors
- **Example:**
 $y_i \sim \mathcal{N}(\theta_i, \sigma^2)$, σ^2 known
 $\theta_i \sim p_0(\theta_i) = \mathcal{N}(\mu_\pi, \sigma_\pi^2)$, μ_π, σ_π^2 unknown

$$\Gamma = \{p : p(\theta) = \prod_{i=1}^M p_0(\theta_i), p_0(\theta_i) = \mathcal{N}(\theta_i | \mu_\pi, \sigma_\pi^2), -\infty < \mu_\pi < \infty, \sigma_\pi^2 > 0\}$$

Priors Close to an Elicited Prior

- We can elicit a single prior p_0 and assume that any prior “close” to p_0 would also be reasonable
- One way to do this is using ϵ -contamination class:

$$\Gamma = \{p : p(\theta) = (1 - \epsilon)p_0(\theta) + \epsilon q(\theta), q \in \mathcal{Q}\}$$

where $0 < \epsilon < 1$ reflects how “close” we feel that p must be to p_0 , and $\mathcal{Q}$ is a class of possible “contaminations”.

3.2 The ML-II Approach to Prior Selection

- As discussed earlier, $p(y)$ reflects the plausibility of $p(y | \theta)$ and $p(\theta)$, in light of the data
- If we treat the sampling model $p(y | \theta)$ as known, $p(y)$ reflects the plausibility of the prior $p(\theta)$.
- Denote the marginal $p(y)$ resulted from prior $\pi(\theta)$ by $m(y | \pi)$
- For observed data y , if $m(y | \pi_1) > m(y | \pi_2)$, we can conclude that the data provides more support for π_1 than for π_2
- Generalizing this idea, we look at $m(y | \pi)$ as a likelihood function for π
- Therefore, a natural method of choosing π is to use maximum likelihood.

Definition 3.1. Suppose Γ is a class of priors under consideration, and that $\hat{\pi} \in \Gamma$ satisfies (for the observed data y)

$$m(y \mid \hat{\pi}) = \sup_{\pi \in \Gamma} m(y \mid \pi)$$

Then $\hat{\pi}$ is called the type II maximum likelihood prior, or ML-II prior for short.

- The determination of an ML-II prior is quite simple for classes with specific functional form. In this case,

$$\sup_{\pi \in \Gamma} m(y \mid \pi) = \sup_{\xi \in \Xi} m(y \mid g(\theta \mid \xi))$$

- So that one simply has to perform a maximization over the hyperparameter ξ

Example. Normal model with known variance

- In our previous example, we had $m(y \mid \pi) = \prod_{i=1}^n m_0(y_i \mid \pi)$, where m_0 is $\mathcal{N}(\mu_\pi, \sigma^2 + \sigma_\pi^2)$
- Therefore,

$$\begin{aligned} m(y \mid \pi) &= \prod_{i=1}^n \frac{1}{[2\pi(\sigma^2 + \sigma_\pi^2)]^{1/2}} \exp \left\{ -\frac{(y_i - \mu_\pi)^2}{2(\sigma^2 + \sigma_\pi^2)} \right\} \\ &= [2\pi(\sigma^2 + \sigma_\pi^2)]^{-n/2} \exp \left\{ -\frac{\sum_{i=1}^n (y_i - \mu_\pi)^2}{2(\sigma^2 + \sigma_\pi^2)} \right\} \\ &= [2\pi(\sigma^2 + \sigma_\pi^2)]^{-n/2} \exp \left\{ \frac{-ns^2}{2(\sigma^2 + \sigma_\pi^2)} \right\} \exp \left\{ \frac{-n(\bar{y} - \mu_\pi)^2}{2(\sigma^2 + \sigma_\pi^2)} \right\} \end{aligned}$$

where $\bar{y} = \sum_{i=1}^n y_i/n$, and $s^2 = \sum_{i=1}^n (y_i - \bar{y})^2/n$

- We seek to maximize $m(y \mid \pi)$ over hyperparameters μ_π and σ_π^2
- The maximum over μ_π is attained at $\bar{y}$, regardless of the value of σ_π^2
That is, $\hat{\mu}_\pi = \bar{y}$ will always be the ML-II choice of μ_π
- Inserting this value into the expression for $m(y \mid \pi)$, it remains to maximize

$$\psi(\sigma_\pi^2) = [2\pi(\sigma^2 + \sigma_\pi^2)]^{-n/2} \exp \left\{ \frac{-ns^2}{2(\sigma^2 + \sigma_\pi^2)} \right\}$$

over σ_π^2 .

- As usual, it is easier to maximize $\log(\psi(\sigma_\pi^2))$

- We have

$$\frac{d}{d\sigma_\pi^2} \log(\psi(\sigma_\pi^2)) = \frac{-n/2}{(\sigma^2 + \sigma_\pi^2)} + \frac{ns^2}{2(\sigma^2 + \sigma_\pi^2)^2}$$

Setting it equal to zero, this yields the maximum at $\sigma_\pi^2 = s^2 - \sigma^2$, provided that $s^2 \geq \sigma^2$. If $s^2 < \sigma^2$, the derivative is always negative, so that the maximum is achieved at $\sigma_\pi^2 = 0$.

- This we have the ML-II estimate of σ_π^2 equal to $\max\{0, s^2 - \sigma^2\}$
- In conclusion, the ML-II prior, $\hat{\pi}$, is

$$\mathcal{N}(\hat{\mu}_\pi, \hat{\sigma}_\pi^2), \quad \text{where} \quad \hat{\mu}_\pi = \bar{y} \quad \text{and} \quad \hat{\sigma}_\pi^2 = \max\{0, s^2 - \sigma^2\}.$$